

Chapter 10

Matrices and Gaussian Elimination

Last week, we solved systems by writing and rewriting equations. That method is honest, but the variable names x, y, z do not really change while we work. What moves are the coefficients. A matrix keeps exactly the information that changes and places it in a grid where the bookkeeping is easier to control.

Solving a system by elimination, the variable names x, y, z are carried along unchanged through every step; only the coefficients move. A matrix strips away the names and keeps the numbers in a grid, so the bookkeeping is done on coefficients alone. The elimination procedure then becomes a fixed sequence of row operations on that grid, and the method, applied systematically, is Gaussian elimination.

10.1 The Augmented Matrix

A linear system is summarized by its coefficients and right-hand sides, arranged in a rectangular array. Each row is an equation; each column before the bar is a variable; the column after the bar holds the constants. The vertical bar is not a new operation; it is only a reminder that the right-hand side has been placed next to the coefficients.

Definition: Augmented matrix. The *augmented matrix* of a linear system records each equation's coefficients in a row, aligned by variable in columns, with the constant terms placed after a vertical bar. The system

$$\begin{array}{rcl} 2x + y - z = 1 \\ x - y + 2z = 3 \\ 3x + 2y + z = 10 \end{array} \quad \text{becomes} \quad \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 1 & -1 & 2 & 3 \\ 3 & 2 & 1 & 10 \end{array} \right].$$

Key idea. The augmented matrix is pure bookkeeping: the same system with the variable names removed. Every operation legal on equations corresponds to an operation on rows, and nothing is lost, because the column positions remember which variable each number belongs to.

For example, the first column above belongs to x , the second to y , and the third to z . A zero coefficient would still need to be written as a zero in its column; otherwise the alignment of the variables would be lost.

10.2 Row Operations

The three reversible moves on equations, from the previous chapter, become the three elementary row operations. Each leaves the solution set unchanged. The guiding rule is simple: a row is an equation, so a legal row move must be a legal equation move.

Key idea. Elementary row operations.

1. *Swap* two rows (reorder the equations).
2. *Scale* a row by a nonzero constant (multiply an equation through).
3. *Replace* a row by itself plus a multiple of another row (add a multiple of one equation to another).

The goal of these operations is *row echelon form*: a staircase pattern in which each row's first nonzero entry (its leading entry) sits to the right of the one above, and entries below each leading position are zero. A matrix in this form is the matrix version of a triangular system, ready for back-substitution.

The picture. Row echelon form is a staircase of zeros below the diagonal. The leading entry of each row steps rightward as you move down, and everything beneath a leading entry is cleared to zero. Reading the bottom row gives one variable; reading upward recovers the rest.

This is the same triangular shape as before, but compressed into a table of numbers. The work is not conceptually different from elimination by hand; the notation has just become more efficient.

10.3 Gaussian Elimination

Gaussian elimination is the disciplined use of row operations to reach echelon form, working one column at a time from left to right: use the leading entry of a row to clear all entries below it, then move to the next column. Once the staircase is complete, back-substitution finishes the solve.

The method is deliberately mechanical. At each stage, choose a leading entry, use it to create zeros below it, and then move down and to the right. This regularity is what makes the method reliable, especially when a system is too large to solve comfortably by informal substitution.

Example 10.3.1. (*a full elimination*) Solve

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array} \right].$$

Clear the first column below the leading 1. Replace row 2 by row 2 minus twice row 1, and row 3 by row 3 minus row 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array} \right].$$

Clear the second column below the new leading entry. Swap rows 2 and 3 for a tidy leading 1, then replace row 3 by row 3 plus three times row 2:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & -1 & -9 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array} \right].$$

The matrix is in echelon form. The bottom row reads $-7z = -21$, so $z = 3$. The middle row reads $y - 2z = -4$, so $y = 2$. The top row reads $x + y + z = 6$, so $x = 1$. The solution is $(1, 2, 3)$, matching the same system solved by hand in the previous chapter.

The row operations did not solve the system all at once. They prepared the system so that the solution could be read from the bottom upward. This is why echelon form is the real target of the method.

Key idea. Gaussian elimination reaches echelon form column by column: pick a leading entry, clear everything below it, advance. The procedure is mechanical and always terminates, which is why it scales to systems far larger than hand elimination can manage.

10.4 Reading the Outcome from the Matrix

The three outcomes appear in the echelon form as clearly as they did in the geometry. A full staircase with a leading entry in every variable column gives a unique solution. A row of zeros in the coefficients with a nonzero constant, the pattern $[0 \ 0 \ 0 \mid c]$ with $c \neq 0$, reads $0 = c$, a contradiction, so no solution. A row of all zeros, $[0 \ 0 \ 0 \mid 0]$, reads $0 = 0$, carrying no information and leaving a free variable, so infinitely many.

Key idea. Reading echelon form.

- A leading entry in every variable column: *unique solution*.
- A row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$: *no solution* (the false statement $0 = c$).

- A row of all zeros and a column without a leading entry: *infinitely many*, with a free variable.

The bottom row is often the most revealing row. It either gives the last variable, exposes a contradiction, or disappears into the identity $0 = 0$.

Example 10.4.1. (*a system with no solution*) Eliminating

$$\left[\begin{array}{cc|c} 1 & -1 & 2 \\ 2 & -2 & 7 \end{array} \right]$$

by replacing row 2 with row 2 minus twice row 1 gives

$$\left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & 0 & 3 \end{array} \right].$$

The bottom row reads $0 = 3$, false, so the system has no solution; the two lines are parallel.

Example 10.4.2. (*a system with infinitely many*) Eliminating

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 6 \end{array} \right]$$

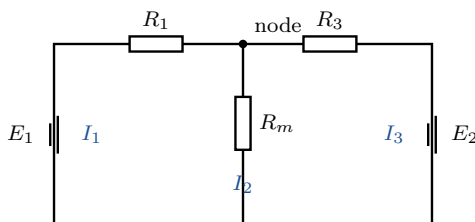
gives a bottom row of all zeros, $0 = 0$. The variable y is free; setting $y = t$, the top row gives $x = 3 - 2t$. The solution set is $(3 - 2t, t)$, the whole line, as before.

10.5 Applications

The applications below use the same method, not a new one. The only new step is the translation: choose variables, write one linear equation for each constraint, then use the matrix method to solve or describe the resulting system.

10.5.1 Currents in a circuit

Kirchhoff's two laws turn a circuit into a linear system. The *current law* states that the current into any junction equals the current out; the *voltage law* states that around any closed loop, the sum of voltage rises from batteries equals the sum of drops across resistors, each drop being resistance times current ($V = IR$). Applying both laws produces one equation per junction and one per loop.



Example 10.5.1. (*a two-loop circuit*) A circuit has a left branch carrying current I_1 through a battery $E_1 = 12$ V and resistor $R_1 = 2\ \Omega$, a middle branch carrying I_2 through $R_m = 3\ \Omega$, and a right branch carrying I_3 through $R_3 = 4\ \Omega$ and battery $E_2 = 10$ V. The current law at the top node and the voltage law around the two loops give

$$\begin{aligned} I_1 - I_2 - I_3 &= 0 && \text{(junction),} \\ 2I_1 + 3I_2 &= 12 && \text{(left loop),} \\ 3I_2 + 4I_3 &= 10 && \text{(right loop).} \end{aligned}$$

The augmented matrix and its elimination are

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 2 & 3 & 0 & 12 \\ 0 & 3 & 4 & 10 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 5 & 2 & 12 \\ 0 & 3 & 4 & 10 \end{array} \right] \xrightarrow{5R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 5 & 2 & 12 \\ 0 & 0 & 14 & 14 \end{array} \right].$$

The bottom row gives $14I_3 = 14$, so $I_3 = 1$ A. Back-substituting, $5I_2 + 2 = 12$ gives $I_2 = 2$ A, and $I_1 - 2 - 1 = 0$ gives $I_1 = 3$ A. The currents are $I_1 = 3$, $I_2 = 2$, $I_3 = 1$ amperes; the junction law checks, since $3 = 2 + 1$.

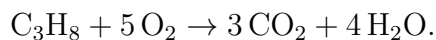
10.5.2 Balancing a chemical equation

Balancing a chemical equation is another conservation problem. The number of atoms of each element must be the same on both sides. Those conservation statements become a linear system for the coefficients in the balanced equation.

Example 10.5.2. (*combustion of propane*) Balance $a\text{C}_3\text{H}_8 + b\text{O}_2 \rightarrow c\text{CO}_2 + d\text{H}_2\text{O}$. Equal atom counts on each side give one equation per element:

$$\begin{aligned} \text{C:} \quad 3a &= c, \\ \text{H:} \quad 8a &= 2d, \\ \text{O:} \quad 2b &= 2c + d. \end{aligned}$$

This is a homogeneous system in a, b, c, d . Take $a = t$ as the free parameter. The carbon equation gives $c = 3t$; the hydrogen equation gives $d = 4t$; substituting into the oxygen equation gives $2b = 2(3t) + 4t = 10t$, so $b = 5t$. The smallest whole numbers come from $t = 1$: $a = 1$, $b = 5$, $c = 3$, $d = 4$, yielding



The free parameter is exactly the freedom to scale a balanced equation up or down.

10.5.3 Flow through a road network

Network-flow problems have the same conservation structure. At each intersection, flow in equals flow out. If the data does not determine every road segment uniquely, the matrix will reveal that with a free parameter.

Example 10.5.3. (*a one-way traffic grid*) Four one-way streets meet at four intersections A, B, C, D , with the unknown segment flows x_1, x_2, x_3, x_4 in cars per hour. Conservation at each intersection (flow in equals flow out), with the external flows shown by the data, gives

$$A : x_1 + x_2 = 800,$$

$$B : x_2 = x_3 + 200,$$

$$C : x_3 + x_4 = 600,$$

$$D : x_1 = x_4.$$

Eliminating leaves one row of zeros: the four conservation equations are not independent, because total flow in must equal total flow out, so one equation follows from the others. The system is underdetermined, with one free parameter. Taking $x_4 = t$, back-substitution gives

$$x_1 = t, \quad x_2 = 800 - t, \quad x_3 = 600 - t, \quad x_4 = t.$$

The flows are not uniquely fixed; the network admits a family of balanced states. The physical requirement that each flow be nonnegative forces $0 \leq t \leq 600$, the range of admissible traffic patterns. A measurement of any one segment pins down all four.

10.6 Exercises

Exercises 10.6

$$3x - y + 2z = 5$$

10.6.1. Write the augmented matrix of $x + 4y - z = 0$, labeling which column

$$2x + y + z = 7$$

belongs to which variable.

10.6.2. State the three elementary row operations and the equation operation each one represents.

10.6.3. Apply one row operation to clear the first column below the top entry of $\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 2 & 1 & 4 \end{array} \right]$, and write the result.

10.6.4. Solve by Gaussian elimination: $\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 1 & 5 \end{array} \right]$, and state the solution as a

point.

10.6.5. Carry $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 1 & -1 & 3 \\ 1 & -2 & 3 & 11 \end{array} \right]$ to echelon form and solve by back-substitution; check your answer.

10.6.6. Eliminate $\left[\begin{array}{cc|c} 1 & -2 & 3 \\ -3 & 6 & 5 \end{array} \right]$ and state the outcome, identifying the row that reveals it.

10.6.7. Eliminate $\left[\begin{array}{cc|c} 2 & 4 & 6 \\ 3 & 6 & 9 \end{array} \right]$, identify the free variable, and write the solution set with a parameter.

10.6.8. Explain how each of the three outcomes, unique, none, and infinitely many, appears in the echelon form of the augmented matrix.

10.6.9. A system's echelon form has a bottom row $[0 \ 0 \ 0 \mid 5]$. State the outcome and explain the reasoning in one sentence.

10.6.10. Balance $a \text{ Fe} + b \text{ O}_2 \rightarrow c \text{ Fe}_2\text{O}_3$ by writing the atom-count equations as a linear system and choosing the smallest whole-number solution.

$$I_1 - I_2 - I_3 = 0$$

10.6.11. A two-loop circuit gives the system $I_1 + 2I_2 = 10$ for its branch currents. Solve by Gaussian elimination, and verify the junction law $I_1 = I_2 + I_3$.

$$2I_2 + 3I_3 = 9$$

10.6.12. Explain why Gaussian elimination on a matrix and hand elimination on the equations produce the same solution, and why the matrix form is preferred for large systems.